

# **A User-centered Evaluation and Refinement of the Critical Code Reviewer (CCR)**

## **Framework**

### **1. Project Overview**

Educational software increasingly incorporates automated and AI-driven decision-making to personalize learning, provide feedback, and support educational experiences. As these technologies become more prevalent, software developers face growing responsibility to ensure that educational systems are fair, transparent, accessible, trustworthy, and inclusive.

Educational contexts raise distinctive ethical demands. Unlike consumer software, educational systems are often predetermined by institutions rather than learners, operate within institutional power asymmetries between learners and the organizations that assess them, and produce outputs whose effects can compound across a learner's trajectory. General-purpose responsible-AI guidance does not address these context-specific stakes.

At the same time, the practice of software development is itself changing. Developers increasingly use AI coding assistants to generate, complete, and review code, which introduces new pathways through which bias and exclusion can enter educational software without deliberate design decisions.

The Critical Code Reviewer (CCR) Framework was developed to support ethical code review by providing developers with structured guidance for identifying ethical concerns during software development. The framework includes a CCR Framework Document, which defines ethical review dimensions and criteria, and a CCR Tool (GitHub Action) that operationalizes the framework during code review. The CCR Tool is itself an AI-based system: it uses a large

language model to analyze changed files against the framework and generate structured review feedback.

The framework's current version was derived from a review of scholarly literature. It has not yet been examined by the learners and educators whose experiences it is intended to represent, nor by the developers expected to apply it. This proposal outlines a user-centered validation study designed to evaluate and refine the CCR Framework using feedback from educational technology stakeholders and software developers. The study will assess whether the framework adequately addresses stakeholder concerns while remaining practical and usable during real-world code review.

## **2. Problem Statement**

Ethical software development frameworks, including Value Sensitive Design (Friedman & Hendry, 2019) and ECCOLA (Vakkuri et al., 2021), provide valuable guidance for incorporating ethical considerations into software development. However, these approaches primarily emphasize high-level ethical principles and provide limited practical support for developers during routine code review.

Without feedback by intended users, it remains unclear whether the CCR Framework adequately represents the ethical concerns of educational technology stakeholders or whether software developers perceive the framework as understandable, useful, complete, practical, and usable within existing development workflows. A systematic evaluation is therefore needed to ensure that the framework effectively supports ethical decision-making during educational software development.

## **3. Purpose and Significance of the Study**

The purpose of this study is to evaluate and refine the Critical Code Reviewer (CCR) Framework through feedback from educational technology stakeholders and software developers. The study seeks to identify ethical concerns experienced by educational technology stakeholders when using educational software, including fairness, accessibility, inclusion, transparency, privacy, accountability, and trust. These findings will be used to determine the extent to which the CCR Framework addresses stakeholder concerns and to refine the framework accordingly.

Following framework refinement, software developers will evaluate the revised CCR Framework through sequential think-aloud code review tasks, hybrid card sorting, and retrospective semi-structured interviews. These activities will assess the clarity, usefulness, completeness, practicality, and usability of the framework during code review.

The study makes three contributions. Empirically, it provides an account of ethical concerns in educational software grounded in the experience of those affected rather than in literature alone. Methodologically, it demonstrates a replicable procedure for evaluating and refining practitioner-facing ethics frameworks. Practically, it produces a refined framework and tool for use in educational software development, including software developed with AI assistance.

## **4. Research Questions**

### **Primary Research Questions**

RQ1: What ethical concerns do educational technology stakeholders experience when using educational software, and to what extent do these concerns align with, extend, or challenge the dimensions of the CCR Framework??

### **Secondary Research Questions**

- RQ1a. What ethical, equity, accessibility, inclusion, transparency, privacy, accountability, and trust concerns do stakeholders describe from their own experience, including experiences with AI-enabled educational technologies?
- RQ1b. Which of these concerns are represented within the current CCR Framework, and which are absent or insufficiently addressed?
- RQ1c. How do stakeholders assess the relevance and importance of the framework's dimensions when these are presented to them?

RQ2: How do software developers experience the revised CCR Framework and CCR Tool when applying them to a code review task?

## **Secondary Research Questions**

- RQ2a. How do developers conceptually organize the framework's review criteria, and to what extent does their organization correspond to the framework's dimensional structure?
- RQ2b. How do developers perceive the clarity, usefulness, completeness, and practicality of the CCR Framework Document during code review?
- RQ2c. How do developers perceive the usability of the CCR Tool, and how do they interpret and evaluate its AI-generated review feedback?

## **5. Operational Definitions**

### **1. CCR) Framework**

The complete ethical code review approach consists of the CCR Framework Document and the CCR GitHub Action Tool.

2. **CCR Framework Document:** The document containing the ethical review dimensions, definitions, and review criteria.

3. **CCR Tool:** The command-line interface (CLI) implementation that operationalizes the CCR Framework by using a large language model to analyze source code and generate structured ethical code review feedback.

**4. Educational Technology Stakeholders**

Individuals who use or are affected by educational software, including students, instructors, and educational technology practitioners.

**5. Software Developers**

Individuals with professional or substantial academic software development experience who have participated in peer code review.

**6. Ethical Concerns**

Issues related to fairness, accessibility, transparency, privacy, accountability, trust, and responsible AI that participants experience when using educational software.

**7. Framework Document Clarity**

The extent to which participants understand the CCR Framework, its dimensions, and its terminology without additional explanation

**8. Framework Document Usefulness**

The extent to which participants believe the CCR Framework helps identify ethical issues in educational software.

## **9. Framework Document Completeness**

The extent to which participants believe the CCR Framework covers the ethical concerns that are important in educational software.

## **10. Framework Document Practicality**

The extent to which software developers believe the CCR Framework can be applied during software development and code review.

## **11. Conceptual Organization**

The extent to which the framework's grouping of criteria into dimensions corresponds to participants' own mental models.

## **12. Usability**

The extent to which developers can use the CCR Tool to accomplish a code review task with effectiveness, efficiency, and satisfaction (ISO 9241-210, 2019)

## **Methodology**

### **Study Design**

This study will employ a two-phase sequential qualitative design to support the user-centered evaluation and iterative refinement of the Critical Code Reviewer (CCR) Framework. Qualitative findings from each phase will inform the development of a subsequent version of the framework. Multiple qualitative data sources, including focus group discussions, think-aloud verbalizations, hybrid card sorting, retrospective interviews, and researcher field notes, will be integrated to support the iterative refinement of the framework.

The study will be conducted in two sequential phases. In Phase 1, educational technology stakeholders will participate in focus group discussions designed to identify ethical concerns related to fairness, accessibility, inclusion, transparency, privacy, accountability, trust, and other issues associated with educational software. Following the focus group discussion, participants will complete a framework-reaction activity in which they evaluate the relevance and completeness of the CCR Framework through plain-language scenarios. Findings from this phase will inform revisions to the conceptual layer of the framework, resulting in Version 2.0 of the CCR Framework.

In Phase 2, software developers will evaluate Version 2.0 of the CCR Framework through counterbalanced code review tasks using a think-aloud protocol, a hybrid card sorting activity, and a retrospective semi-structured interview. These activities are designed to evaluate how developers interpret, organize, and apply the framework during authentic code review while identifying opportunities to improve the framework's conceptual organization, clarity, completeness, practicality, and operational implementation within the CCR Tool.

The sequential design enables findings from Phase 1 to inform the refinement of the framework before it is evaluated by software developers in Phase 2. Guided by user-centered design principles (Gould & Lewis, 1985; ISO 9241-210, 2019), the study integrates multiple qualitative data sources to support the iterative evaluation and refinement of the Critical Code Reviewer (CCR) Framework.

### **Layers of Revision**

The framework has a conceptual layer (dimensions, definitions, terminology) and an operational layer (criterion indicators, review levels, detection logic). Phase 1 participants are positioned to inform the conceptual layer; Phase 2 participants inform both. The research team is responsible for translating conceptual revisions into operational changes, documented in the change log.

### **Version control**

Version 1.0 (literature-derived) → Phase 1 → Version 2.0 (stakeholder-informed) → Phase 2 → Version 3.0 (developer-refined). Because Version 3.0 is produced after data collection concludes, it is presented as a proposed revision. A confirmatory member reflection round provides limited evaluation of Version 3.0; fuller evaluation is identified as future work.

### **Tool synchronization (gating milestone)**

Before Phase 2 begins, the CCR Tool's prompt architecture, output, and test suite will be updated to reflect Framework Version 2.0 and verified to produce output consistent with the revised dimensions. Phase 2 will not commence until this verification is documented.

### **Phase 1: Educational Technology Stakeholder Evaluation ( 60 Minutes)**

The purpose of Phase 1 is to identify the ethical concerns experienced by educational technology stakeholders when using educational software and to evaluate the extent to which these concerns

are represented within the current version of the Critical Code Reviewer (CCR) Framework.

Findings from this phase will be used to refine the conceptual layer of the framework, including its dimensions, definitions, terminology, and overall organization, resulting in **Version 2.0** of the CCR Framework.

Approximately 12 participants will be organized into three to four focus groups, each consisting of three to four participants. Conducting the study as focus groups allows participants to build upon one another's experiences, compare perspectives, and collectively identify ethical concerns that may not emerge during individual interviews. The focus group format also provides a more efficient approach to collecting diverse stakeholder perspectives while supporting discussion around shared experiences with educational software.

Each focus group session will consist of three sequential activities:

1. Demographic Questionnaire
2. Focus Group Discussion
3. Framework Reaction Activity

The activities are designed to ensure that participants first discuss their own experiences with educational software before being introduced to the CCR Framework. This sequence minimizes the possibility that the framework influences participants' descriptions of ethical concerns.

### **Activity 1: Demographic Questionnaire ( 5 minutes )**

The session will begin with a brief demographic questionnaire designed to collect background information about participants' educational roles, experience with educational software, and

familiarity with AI-enabled educational technologies. The questionnaire will provide contextual information to support interpretation of participants' responses and describe the diversity of the participant sample.

### **Activity 2: Focus Group Discussion ( 30 minutes )**

The session will continue with a moderated focus group discussion designed to explore participants' experiences using educational software, including AI-enabled educational technologies such as automated grading systems, adaptive learning platforms, AI tutors and chatbots, and automated proctoring systems. Participants will discuss ethical, accessibility, inclusion, equity, transparency, privacy, accountability, and trust concerns they have experienced while interacting with educational technologies.

Discussion prompts will be informed by the Critical Incident Technique (Flanagan, 1954). At the beginning of the discussion, each participant will be invited to describe a specific experience involving educational software before the discussion is opened to the group. This approach encourages participants to share concrete experiences while allowing subsequent discussion to build upon shared and contrasting perspectives.

Throughout the discussion, participants will also be encouraged to describe challenges they have encountered, explain why these issues were important to them, and recommend ways educational software could better support ethical and inclusive learning experiences.

With participants' consent, the discussion will be audio-recorded, and the researcher will maintain observational field notes documenting key interactions, areas of agreement and disagreement, and notable group dynamics.

### **Activity 3: Framework Reaction ( 25 minutes )**

Following the focus group discussion, participants will be presented with the dimensions of the CCR Framework through plain-language scenarios. Because participants are not expected to read or interpret source code, each framework dimension will be represented by one or more concrete vignettes describing observable situations rather than technical review criteria. For example, scenarios may describe a registration form that rejects names containing accents or apostrophes or an automated risk flag that follows a student throughout an academic year without reassessment.

Each scenario will be presented to the group, and participants will discuss whether it reflects their experiences, how important they believe the issue is, and whether they have encountered similar situations while using educational software. The facilitator will encourage all participants to contribute to the discussion and will probe for differing viewpoints when appropriate. After all scenarios have been discussed, participants will be invited to identify ethical concerns they believe are missing from the framework and suggest additional considerations that software developers should address when developing educational software.

### **Phase 1 Outcome**

Following completion of Phase 1, focus group discussions and framework-reaction activities will be analyzed using Template Analysis (King, 2012; King & Brooks, 2017). . The analysis will begin with the existing CCR Framework as the initial coding template and will be iteratively refined based on participants' discussions and feedback. The refined coding template will then be compared with the existing framework to determine how well the framework represents stakeholders' identified ethical concerns and where revisions are needed.

Findings from this phase will inform revisions to the framework's conceptual layer, including refinement of existing dimensions, clarification of terminology, modification of review criteria, and the addition of new ethical considerations where appropriate. These refinements will result in Version 2.0 of the Critical Code Reviewer (CCR) Framework, which will subsequently be evaluated by software developers in Phase 2.

### **Phase 2: Software Developer Validation ( Approximately 100 minutes )**

The purpose of Phase 2 is to evaluate software developers' experiences using Version 2.0 of the Critical Code Reviewer (CCR) Framework during authentic code review activities. This phase seeks to understand how developers interpret and apply the framework, how the framework influences their code review process, and how they perceive the clarity, conceptual organization, completeness, practicality, and usability of both the CCR Framework Document and the CCR Tool. Findings from this phase will inform the refinement of both the conceptual and operational components of the framework, resulting in Version 3.0 of the CCR Framework.

Each participant will complete an individual session lasting approximately 100 minutes. The session will consist of five sequential activities:

1. A demographic questionnaire.
2. Sequential Code Review Tasks
3. A hybrid card sorting activity.
4. A retrospective semi-structured interview

The sequence is designed to ensure that participants first experience the framework in an authentic code review context before evaluating its organization, usability, and overall effectiveness.

### **Activity 1: Demographic Questionnaire ( 10 minutes )**

The session will begin with a brief demographic questionnaire designed to collect background information about participants' software development experience, programming background, familiarity with code review practices, and experience using collaborative development platforms and AI-assisted software development tools. The questionnaire will provide contextual information for describing the participant sample and interpreting participants' experiences during the study.

### **Activity 2: Sequential Code Review Tasks (50 minutes )**

Participants will complete two sequential code review tasks using two functionally equivalent pull requests derived from a standardized educational software code base developed specifically for the study. Using a standardized code base ensures that all participants evaluate comparable ethical issues while minimizing variability introduced by differences in software architecture or application domain. Prior to data collection, both pull requests will be reviewed and pilot-tested by the research team to ensure comparable scope, complexity, and review effort.

#### **Task 1: Independent Code Review**

Participants will first complete an independent code review using their usual code review practices without access to the CCR Framework Document or the CCR Tool. During the review, participants will be asked to think aloud by continuously verbalizing their observations,

interpretations, and decision-making processes. This task is intended to establish participants' existing code review practices before they are introduced to the CCR Framework.

### **Task 2: CCR-Supported Code Review**

Following completion of the independent code review, participants will complete a second code review using **Version 2.0 of the CCR Framework Document** and the **CCR Tool (CLI)**.

Participants will again think aloud while applying the framework and tool throughout the review process. The researcher will encourage participants to continue verbalizing their thoughts when necessary and may ask brief, non-leading follow-up questions to clarify participants' reasoning without influencing their review decisions.

The independent code review provides insight into participants' existing code review practices, whereas the CCR-supported code review examines how the CCR Framework and CCR Tool influence developers' reasoning, identification of ethical concerns, and code review decisions.

Throughout both tasks, the researcher will maintain observational field notes documenting participants' review strategies, interactions with the framework and tool, notable behaviors, areas of confusion, and other observations relevant to framework refinement.

### **Activity 3: Hybrid Card Sorting (20 minutes )**

Following completion of both code review tasks, participants will complete a hybrid card sorting activity using Version 2.0 of the CCR Framework Document. Conducting the card sorting activity after the code reviews allows participants to evaluate the framework based on practical experience rather than initial impressions.

Using FigJam, participants will organize review criteria under the framework dimensions they believe best represent them. Participants may move criteria between dimensions, identify overlapping or redundant dimensions, recommend changes to terminology, suggest additional dimensions or review criteria, and explain the reasoning behind each organizational decision. The researcher will observe participants' interactions with the framework and document field notes describing notable behaviors, areas of uncertainty, and participant comments.

#### **Activity 4: Retrospective Framework Interview (20 minutes )**

Finally, participants will take part in a semi-structured retrospective interview. They will be asked to compare their experiences reviewing code with and without CCR support, discuss how the framework influenced their review process, identify dimensions or criteria they found useful or confusing, describe missing ethical considerations, comment on the usefulness of the AI-generated feedback, and recommend improvements to both the CCR Framework Document and the CCR Tool.

#### **Phase 2 Outcome**

Following completion of Phase 2, data from the sequential code review tasks, think-aloud verbalizations, hybrid card sorting activities, retrospective interviews, and researcher field notes will be analyzed using Template Analysis (King, 2012; King & Brooks, 2017). Building on the Version 2.0 coding template developed during Phase 1, the analysis will examine software developers' experiences applying the CCR Framework and CCR Tool during authentic code review activities. Particular attention will be given to identifying framework dimensions and review criteria that participants found clear, useful, practical, complete, unclear, redundant, or difficult to apply, as well as recommendations for improving the operational implementation of the CCR Tool.

Using the predefined framework refinement decision rules, findings from Phase 2 will inform revisions to both the conceptual and operational layers of the framework. These revisions may include clarifying terminology, reorganizing framework dimensions, modifying review criteria, improving the AI-generated review process, and refining the interaction between the CCR Framework Document and the CCR Tool.

The resulting revisions will produce Version 3.0 of the Critical Code Reviewer (CCR) Framework, representing the final refined version of the framework developed through the integration of educational technology stakeholder perspectives and software developer evaluations. A summary of all framework revisions and the rationale supporting each change will be documented in an audit trail to ensure transparency throughout the refinement process.

## **Participants**

This study will involve two purposively selected participant groups recruited across two sequential phases. Phase 1 will involve educational technology stakeholders whose experiences with educational software will inform the refinement of the Critical Code Reviewer (CCR) Framework. Phase 2 will involve software developers who will evaluate the revised CCR Framework and CCR Tool during authentic code review activities. Purposive sampling will be used because qualitative research seeks information-rich participants capable of providing detailed insight into the phenomenon under investigation (Patton, 2015).

### **Phase 1: Educational Technology Stakeholders**

Approximately 12 educational technology stakeholders will be recruited using maximum variation purposive sampling to capture diverse perspectives on the ethical use of educational

software. Organizing participants into three to four small focus groups is consistent with recommendations for applied qualitative research (Krueger & Casey, 2015).

**Participants may include:**

- Undergraduate students
- Graduate students
- Instructors
- Educational technology practitioners
- Instructional designers

The proposed sample size is guided by the principle of information power (Malterud et al., 2016), which suggests that qualitative studies with focused aims and information-rich participants require relatively small sample sizes. Consequently, participants will be purposefully selected based on their ability to provide rich, relevant, and diverse experiences. Recruitment will continue until sufficient information power has been achieved to adequately address the research questions.

**Inclusion Criteria**

Participants must:

- Be at least 18 years of age
- Have experience using educational software;
- Provide informed consent.

Participants with experience using educational technologies, such as intelligent tutoring systems, adaptive learning platforms, AI chatbots, automated grading systems, or automated proctoring systems, will be particularly valuable because these technologies introduce ethical considerations directly relevant to the study.

### **Exclusion Criteria**

Individuals will be excluded if they:

- Are younger than 18 years of age
- Have no prior experience using educational software;
- Are unwilling to provide informed consent.

### **Phase 2: Software Developers**

Approximately 12 software developers will be recruited using purposive sampling. Participants will be selected because they possess the knowledge and practical experience necessary to evaluate the CCR Framework Document and CCR Tool during authentic code review activities.

#### **Potential participants may include:**

- Professional software developers
- Software engineers
- Graduate students with substantial software development experience
- Researchers who regularly participate in peer code reviews

The proposed sample size is also guided by the principle of information power (Malterud et al., 2016). The study has a narrowly defined objective of evaluating and refining the CCR Framework and CCR Tool for ethical code review in educational software. Participants will therefore be purposefully recruited from an information-rich population with direct experience in software development and peer code review. Recruitment will continue until the research team determines that the collected data provide sufficient information power to address the research questions.

### **Inclusion Criteria**

Participants must:

- Be 18 years of age or older.
- Have experience with software development or code review.
- Have experience reviewing code using platforms such as GitHub, or comparable code review systems.
- Be able to provide informed consent.
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Eligibility will be confirmed through a brief screening questionnaire administered before enrollment.

### **Exclusion Criteria**

- Individuals under the age of 18 years.
- Adults unable to provide informed consent due to impaired decision-making capacity.
- Have no prior experience conducting peer code reviews.

- Individuals who do not meet the eligibility criteria described above.

## **Data Analysis Plan**

Qualitative data will be analyzed using Template Analysis (King, 2012; King & Brooks, 2017), a flexible approach that is particularly appropriate for studies that begin with an existing conceptual framework while remaining open to new concepts emerging from participant data.

The initial coding template will be based on the dimensions and review criteria of Version 1.0 of the Critical Code Reviewer (CCR) Framework. Throughout the analysis, the template will be iteratively refined to reflect participants' feedback by confirming, modifying, merging, splitting, relocating, renaming, or adding framework dimensions and review criteria. This approach aligns with the study's objective of iteratively refining an existing framework rather than developing an entirely new theoretical model.

## **Phase 1: Educational Technology Stakeholder Data**

### **Focus Group**

Audio recordings from the focus group discussions will be transcribed verbatim before analysis.

Focus group transcripts will be analyzed using Template Analysis (King, 2012; King & Brooks, 2017).

The analysis will begin with an initial coding template derived from the existing CCR Framework, including its dimensions and review criteria. Researchers will code participants' descriptions of ethical concerns, priorities, expectations, and recommendations related to educational software and AI-enabled educational technologies. As coding progresses, the template will be iteratively refined to accommodate stakeholder perspectives that are not adequately represented by the existing framework.

The analysis will specifically examine:

- Ethical concerns that align with existing CCR dimensions.
- Concerns that require modification of existing review criteria.
- Concerns suggesting the need to merge, split, relocate, or rename framework dimensions or criteria.
- New ethical concerns requiring additional dimensions or review criteria.

Special attention will be given to issues involving fairness, accessibility, inclusion, transparency, privacy, accountability, trust, and responsible AI.

### **Framework Crosswalk**

The resulting coded findings and refined coding template will be compared with the existing dimensions and review criteria of the Critical Code Reviewer (CCR) Framework to identify:

- ethical concerns already represented within the framework;
- concerns that are missing or insufficiently addressed;
- opportunities to refine framework terminology;
- opportunities to revise, merge, or expand framework dimensions.

Framework-reaction responses will be analyzed alongside the crosswalk to provide participant-level evidence of coverage and importance.

### **Phase 2: Software Developer Data**

Qualitative data collected during Phase 2 will include demographic questionnaires, think-aloud verbalizations recorded during the code review activities, hybrid card sorting artifacts, retrospective semi-structured interviews, and researcher field notes. All audio recordings will be

transcribed verbatim prior to analysis. Consistent with Phase 1, these data will be analyzed using Template Analysis (King, 2012; King & Brooks, 2017). The initial coding template will be based on Version 2.0 of the Critical Code Reviewer (CCR) Framework, which was refined following Phase 1. Throughout the analysis, the coding template will be iteratively modified to reflect participants' experiences, interpretations, and recommendations for improving both the CCR Framework and the CCR Tool.

### **Code Review and Think-Aloud Analysis**

Audio recordings from the sequential code review activities and concurrent think-aloud verbalizations will be transcribed verbatim prior to analysis. Data from both the independent code review and the CCR-supported code review will be analyzed using Template Analysis (King, 2012; King & Brooks, 2017). The initial coding template will be based on Version 2.0 of the Critical Code Reviewer (CCR) Framework, while remaining open to modifications that emerge from participants' experiences.

The analysis will focus on understanding how software developers interpret, apply, and evaluate the CCR Framework during authentic code review activities. Particular attention will be given to:

- interpretation of framework dimensions and review criteria;
- application of the framework during code review;
- ethical issues identified during the review process;
- decision-making processes while reviewing code;
- challenges encountered when applying the framework;
- recommendations for improving the framework; and
- recommendations for improving the CCR Tool.

Researcher field notes documenting participant behaviors, moments of hesitation, requests for clarification, and other contextual observations will be analyzed alongside the transcripts to support interpretation of participants' experiences.

### **Hybrid Card Sorting Analysis**

Data from the hybrid card sorting activity will be analyzed using Template Analysis (King, 2012; King & Brooks, 2017). to examine how software developers conceptualize the organization of the revised CCR Framework. Participants' card arrangements, verbal explanations, and researcher observations will be analyzed together to understand how developers organize, interpret, and relate the framework's dimensions and review criteria.

The analysis will examine:

- Agreement in how participants organized review criteria within framework dimensions.
- Review criteria consistently relocated to different dimensions.
- Dimensions perceived as overlapping or conceptually redundant.
- Review criteria participants found unclear or difficult to interpret.
- Missing review criteria or framework dimensions identified by participants.
- Newly proposed dimensions, review criteria, or organizational structures.

These findings will be used to identify opportunities to improve the conceptual organization, terminology, and completeness of the CCR Framework.

## **Retrospective Interview Analysis**

Audio recordings from the retrospective semi-structured interviews will be transcribed verbatim and analyzed using the evolving coding template. The interviews will provide participants with an opportunity to reflect on their overall experience using the CCR Framework and the CCR Tool after completing the code review and hybrid card sorting activities.

The analysis will explore participants' perceptions of:

- Clarity of the framework's dimensions and review criteria.
- Usefulness of the framework during code review.
- Completeness of the ethical issues addressed by the framework.
- Practicality of integrating the framework into software development workflows.
- Strengths and limitations of the CCR Tool.
- Recommendations for improving both the framework and the supporting tool.

## **Integration of Findings**

Following the separate analyses of Phase 1 and Phase 2, findings will be integrated to refine **Version 2.0** of the Critical Code Reviewer (CCR) Framework and produce **Version 3.0**.

Integration will involve synthesizing evidence from:

- Educational technology stakeholder focus group discussions and framework-reaction activities
- Hybrid card sorting
- Think-aloud code review sessions
- Retrospective interviews

- Researcher field notes.

The integrated findings will be examined to identify areas of convergence and divergence across participant groups. Specifically, stakeholder perspectives on ethical priorities and concerns will be compared with software developers' experiences organizing, interpreting, and applying the framework during code review. This process will identify framework dimensions and review criteria that are consistently supported, unclear, redundant, misplaced, or insufficiently represented.

Using the predefined framework refinement decision rules, the research team will determine whether individual dimensions or review criteria should be retained, revised, relocated, merged, split, added, or removed. Each modification will be supported by evidence across the qualitative data sources and documented in an audit trail to ensure transparency throughout the refinement process.

The outcome of this integration will be Version 3.0 of the Critical Code Reviewer (CCR) Framework, representing a refined framework that incorporates educational technology stakeholders' ethical priorities together with software developers' practical experiences using the framework during educational software code review.

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